

Saturated Superfluid Vacuum VII-a

Quantum Mechanics from Hydrodynamics

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Draft

Abstract

This paper develops the quantum-mechanical limit of the Saturated Superfluid Vacuum (SSV) framework. The central idea is that quantum behavior is not a separate foundational axiom, but the linearized or low-amplitude hydrodynamic behavior of the same medium. Quantization is tied to circulation and topology rather than imposed independently, and measurement is treated as an irreversible reconnection process in the medium.

1 Introduction

This paper develops the quantum-mechanical limit of the Saturated Superfluid Vacuum (SSV) framework. The central idea is that quantum behaviour is not a separate foundational axiom, but the low-amplitude hydrodynamic regime of the same medium already used elsewhere in the series. The task here is intentionally narrow: derive the Schrödinger limit from the hydrodynamic variables, interpret uncertainty as a bound on physical localisation in the medium, and treat measurement as an irreversible reconnection event rather than as an extra postulate.

The sections that follow are adapted from the broader legacy SSV V manuscript and gathered here so this paper can stand on its own inside the newer, one-claim-per-paper structure.

2 Quantum Mechanics from Linearised Hydrodynamics

2.1 The Madelung Transformation

Let the superfluid vacuum be described by a complex order parameter $\Psi(\mathbf{x}, t) = \sqrt{\rho(\mathbf{x}, t)} e^{i\varphi(\mathbf{x}, t)}$, where ρ is the local mass density and φ the velocity potential. The velocity field is $\mathbf{v} = (\hbar/m)\nabla\varphi$.

The Gross–Pitaevskii equation (GPE) governing the order parameter is

$$i\hbar\partial_t\Psi = \left[-\frac{\hbar^2}{2m}\nabla^2 + V_{\text{ext}} + g|\Psi|^2 \right] \Psi, \quad (1)$$

where $g = 4\pi\hbar^2 a_s/m$ with a_s the s -wave scattering length and V_{ext} any slowly varying external potential (itself generated by other defects via the mechanisms of Paper II).

Substituting the polar form into (1) and separating real and imaginary parts yields two real equations:

$$\partial_t\rho + \nabla \cdot (\rho\mathbf{v}) = 0 \quad [\text{continuity}], \quad (2)$$

$$m\partial_t\mathbf{v} + m(\mathbf{v} \cdot \nabla)\mathbf{v} = -\nabla(V_{\text{ext}} + g\rho - Q) \quad [\text{Euler}], \quad (3)$$

where

$$Q(\mathbf{x}, t) = -\frac{\hbar^2}{2m} \frac{\nabla^2 \sqrt{\rho}}{\sqrt{\rho}} \quad (4)$$

is the *quantum pressure* (Bohm potential). Equations (2)–(3) are the *Madelung equations*: standard compressible hydrodynamics supplemented by the quantum pressure term Q .

2.2 The Schrödinger Limit

When the density perturbation $\delta\rho = \rho - \rho_0$ is small compared to the saturation density ρ_0 , and spatial gradients are slow on the scale of the vortex core radius $\xi = \hbar/\sqrt{2mg\rho_0}$ (the healing length), the nonlinear interaction term $g|\Psi|^2\Psi$ acts as a uniform background that can be absorbed into a redefined phase, leaving:

$$i\hbar\partial_t\psi = \left[-\frac{\hbar^2}{2m}\nabla^2 + V_{\text{eff}} \right] \psi, \quad (5)$$

where ψ is a slowly varying envelope and V_{eff} encodes the long-range potential produced by all other defects. This is precisely the Schrödinger equation for a particle of mass m in potential V_{eff} .

Key point: Equation (5) is not postulated. It is derived from the single postulate — the saturated superfluid — in the limit where the defect size is small compared to the scales of interest. Quantisation is not an axiom; it follows from the topological quantisation of circulation (Feynman–Onsager relation):

$$\oint_C \mathbf{v} \cdot d\mathbf{l} = \frac{n\hbar}{m}, \quad n \in \mathbb{Z}, \quad (6)$$

which is itself a consequence of the single-valuedness of Ψ around any closed contour encircling a vortex core.

2.3 The Uncertainty Principle as a Hydraulic Bound

The Heisenberg uncertainty relation $\Delta x \Delta p \geq \hbar/2$ follows directly from the Fourier theory of wave packets in Ψ . In the hydrodynamic language it is a statement about the irreducible minimum spread of a density perturbation: a wave packet localised to size Δx must contain momentum modes spread over $\Delta p \sim \hbar/\Delta x$, because the quantum pressure (4) resists compression of the packet below the healing length ξ . There is no need to invoke observer effects or complementarity as primitive concepts; the uncertainty principle is the hydraulic rigidity of the medium.

3 Quantum Measurement as Vortex Reconnection

The measurement problem — why a superposition apparently collapses to a definite outcome upon observation — has no satisfactory resolution within the abstract formalism of quantum mechanics. In SSV it dissolves into a physical process.

3.1 The Physical Picture

A quantum superposition is, in hydrodynamic terms, a coherent linear combination of field configurations in Ψ . As long as the medium remains globally coherent (no irreversible events), interference terms survive and the superposition is maintained. A *measurement* is a physical

interaction that drives the coupled system — probe defect + apparatus — into a regime where vortex reconnection becomes inevitable. Reconnection is topologically irreversible: once two vortex filaments have exchanged segments, the original topology cannot be restored without re-creating the severed strands from zero, which costs an energy of order $\mu_0 = m_e/\alpha \approx 70$ MeV per reconnection event (the flux-tube mass scale derived in Paper I). In practice, macroscopic measurement apparatuses involve $\sim 10^{23}$ such events, making coherence recovery thermodynamically impossible.

3.2 Born’s Rule from Vortex Statistics

The probability of a given reconnection topology being realised — which branch of the superposition is “selected” — is proportional to the squared amplitude of that branch in Ψ , because the energy available to drive reconnection in mode k is proportional to $|\psi_k|^2$. The Born rule

$$P_k = |\langle k|\Psi\rangle|^2 \tag{7}$$

follows from the equiprobability of energy states at reconnection threshold: the mode with larger $|\psi_k|^2$ occupies a larger basin in configuration space and is realised with proportionally higher frequency.

This is a statistical, not a metaphysical, statement — in perfect analogy with the Maxwell–Boltzmann distribution for molecular speeds, which also follows from energy counting without invoking any ontological indeterminism.

4 Focused Predictions

Several near-term consequences of this hydrodynamic reading of quantum mechanics are already implicit in the legacy draft. Two are especially central to the present paper.

- Q1. Vortex-reconnection signal in quantum optics.** Quantum measurement events in optomechanical systems should produce a characteristic acoustic burst in the medium, coincident with the detector click that marks the reconnection event.
- Q2. Sub-percent proton-mass closure from the same hydrodynamic machinery.** The numerical minimisation programme for the proton trefoil breather should tighten the mass prediction to below the current structural error budget, providing a direct test of whether the same hydrodynamic framework really closes both particle structure and the quantum limit.

5 Conclusion

Quantum behavior in SSV is treated here as a hydrodynamic limit of the same underlying medium. Quantisation is tied to circulation and topology, uncertainty to the finite structure of the medium, and measurement to irreversible reconnection rather than to an independent collapse postulate.

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